

§4 一元函数微分学的计算

point •
$$\begin{cases} d(x^n) = nx^{n-1}dx \\ dx^n = (dx)^n \end{cases}$$

eg. 例 4.1 设 $f(x) = (\cos x - 4)\sin x + 3x$.
求 $\frac{df(x)}{dx} = \frac{d(f(x))}{dx} = \frac{1}{2x} f'(x)$.

三. 复合函数求导 [例 4.3 ~ 4.4]

△ $\{f[g(x)]\}' = f'[g(x)] \cdot g'(x)$ ← 对外面求导
△ 注意求导位置: $f'[g(x)] = \frac{d\{f[g(x)]\}}{g(x)}$ ← 对内面求导.

四. 隐函数求导 [例 4.5]

$F(x, y) = 0$. 两边对 x 求导, 注意 $y = y(x)$, y 看作中间变量, 解方程求 y'

五. 反函数求导 [例 4.6]

$y = f(x)$ 可导, 且 $f'(x) \neq 0$. 则有反函数 $x = g(y)$.

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} \Rightarrow \underline{g'(y) = \frac{1}{f'(x)}} \quad \text{—— 倒数}$$

= 链式求导.
$$y_{xx}'' = \frac{d^2 y}{dx^2} = \frac{d(\frac{dy}{dx})}{dx} = \frac{d(\frac{1}{\frac{dx}{dy}})}{dx} = \frac{d(\frac{1}{f'(x)})}{dy} \frac{dy}{dx} = \frac{x_{yy}''}{-(x_y')^2} \cdot \frac{1}{x_y'} = \boxed{\frac{-x_{yy}''}{(x_y')^3}}$$

六. 分段函数求导 (含绝对值). [例 4.7 ~ 4.8]

(1) 在分段点处用导数定义求导 (定义法).

(2) 在非分段点用导数公式求导 (公式法).

七. 多项乘除、开方、乘方 (对数求导法).

step 1 $\ln|y| = \ln|f(x)|$ ← 注意加绝对值
step 2 两边对 x 求导.

$\frac{1}{y} y' = [\ln|f(x)|]' \Rightarrow y' = y [\ln|f(x)|]'$
视绝对值而不见.

八. 幂指函数求导法

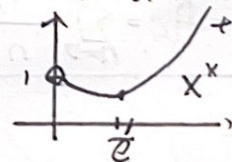
step 1. $u(x)^{v(x)} = e^{v(x) \ln u(x)}$

step 2. 对 x 求导

[注 x^x 图像 → 例 30.1.5.11 $(2+x)e^{\frac{1}{x}}$

$$(x^x)' = (e^{x \ln x})' = x^x \cdot (\ln x + 1) \stackrel{?}{=} 0.$$

$$\Rightarrow x = \frac{1}{e}.$$

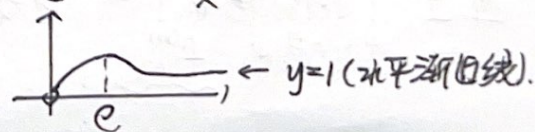


(同 x^{2x} , x^{3x} ...)

$x^{\frac{1}{x}}$ 图像

$$(x^{\frac{1}{x}})' = (e^{\frac{1}{x} \ln x})' = x^{\frac{1}{x}} \cdot \frac{1 - \ln x}{x^2} \stackrel{?}{=} 0.$$

$$\Rightarrow x = e$$



九 参数方程确定的函数求导. [例 4.11]

$$\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases} \quad \left\{ \begin{aligned} \frac{dy}{dx} &= \frac{dy/dt}{dx/dt} = \frac{\psi'(t)}{\varphi'(t)} = \frac{\psi'(t)}{\varphi'(t)} \\ \frac{d^2y}{dx^2} &= \frac{d(\frac{dy}{dx})}{dx} = \frac{\psi''(t)}{\varphi'(t)} \end{aligned} \right. \quad \begin{aligned} &= \text{一阶导} \\ &= \text{二阶导} \end{aligned}$$

十 高阶导数. [例. 4.13, 4.14]

1) 归纳法.

比如. $y = 3^x$ $y' = 3^x \ln 3$ $y'' = 3^x (\ln 3)^2$

2) 莱布尼茨公式. [例 4.15]

$$(uv)^{(n)} = C_n^0 u^{(n)} v + C_n^1 u^{(n-1)} v^{(1)} + \dots + C_n^n u^{(0)} v^{(n)}$$

$$= \sum_{k=0}^n C_n^k u^{(n-k)} v^{(k)}$$

3) 展开式. [例 4.16]

(无穷阶可导): $y = f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} \cdot (x-x_0)^n$

$\downarrow x_0=0$

$$y = f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

通过 [泰勒展开] 与泰勒级数

$\Rightarrow$ 比较得出 $f^{(n)}(x_0)$

例4.2 ★

设 $f'(0)=1$, $f''(0)=0$. 证明在 $x=0$ 处, 有 $\frac{d^2}{dx^2} f(x) = \frac{d^2}{dx^2} f^2(x)$

[分析]

$$\left(\frac{d^2 y}{dx^2} = \frac{d(\frac{dy}{dx})}{dx} = \frac{d^2 y}{(dx)^2} = y'' \right)$$

记 $F(x) = f(x)$

$$\frac{d^2}{dx^2} f(x) \Big|_{x=0} = F''(0) = \lim_{x \rightarrow 0} \frac{F'(x) - F'(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f'(x) - 1}{x}$$

$$= 2 \lim_{x \rightarrow 0} f'(x) = 2$$

$\therefore f''(0)=0 \Rightarrow f'(x)$ 在 $x=0$ 可导.

记 $G(x) = f^2(x)$

$$\frac{d^2}{dx^2} f^2(x) \Big|_{x=0}$$

$$G'(0) = \lim_{x \rightarrow 0} \frac{G(x) - G(0)}{x - 0}$$

$$= \lim_{x \rightarrow 0} \frac{2f(x) \cdot f'(x) - 2f(0)f'(0)}{x}$$

$$(\text{重要导数定义}) = 2 \lim_{x \rightarrow 0} \frac{f(x) \cdot f'(x) - f(x) + f(x) - f(0)}{x}$$

$$= 2 \lim_{x \rightarrow 0} f(x) \left[\frac{f'(x) - 1}{x} \right] + \lim_{x \rightarrow 0} 2 \frac{f(x) - f(0)}{x}$$

= 导数定义

- 导数定义

$$= 2f(0)f''(0) + 2f'(0) = 2$$

例4.3

设 $y = \ln^2(\sin^2 x + 1)$ 求 y'

$$y' = 2 \ln^2(\sin^2 x + 1) \cdot \frac{2 \sin x}{\sin^2 x + 1} \cdot \cos x$$

例4.4

设 $g(x)$ 可微, $h(x) = e^{1+g(x)}$, $h'(1)=1$, $g'(1)=2$, 求 $g(1)=?$

$$h(x) = e^{1+g(x)} \quad h'(x) = e^{1+g(x)} \cdot g'(x)$$

$$h'(1) = e^{1+g(1)} \cdot g'(1) \Rightarrow 1 = 2e^{1+g(1)} \Rightarrow g(1) = -\ln 2 - 1$$

例4.5

设 $y=y(x)$ 隐函数 $\rightarrow x^2 - y + 1 = e^y$, $\frac{dy}{dx} \Big|_{x=0} = ?$

$$x^2 - y + 1 = e^y \quad \xrightarrow{x=0} \quad 1 - y = e^y \Rightarrow y=0$$

$$2x - y' = e^y \cdot y' \quad \xrightarrow{x=0} \quad y' = 0$$

$$2 - y'' = e^y (y')^2 + e^y y'' \quad \xrightarrow{x=0} \quad y'' = 1$$

例4.6

设 $y=f(x)$ 与 $x=g(y)$ 互为反函数, $y=f(x)$ 可导,

且 $f'(x) \neq 0$, $f(3)=5$, $h(x) = f[\frac{1}{3}g^2(x^2+3x+1)]$ 求 $h'(1)$.

[分析] $h'(x) = f'[\frac{1}{3}g^2(x^2+3x+1)] \cdot \frac{2}{3}g(x^2+3x+1) \cdot g'(x^2+3x+1) \cdot (2x+3)$

$$h'(1) = f'[\frac{1}{3}g^2(5)] \cdot \frac{2}{3}g(5) \cdot g'(5) \cdot 5$$

$$f(3)=5 \Rightarrow g(5)=3$$

$$h'(1) = f'(3) \cdot g'(5) \cdot 10 \quad \text{互为倒数!}$$

$$= 10$$

例4.7

设 $f(x)$ 在 $(-\infty, +\infty)$ 内连续且大于 0

$$g(x) = \begin{cases} \frac{\int_0^x t f(t) dt}{\int_0^x f(t) dt}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

(1) 求 $g'(x)$

(2) 证明 $g'(x)$ 在 $(-\infty, +\infty)$ 内连续.

[分析] $x \neq 0$

$$g'(x) = \frac{f(x) \int_0^x f(t) dt - f(x) \int_0^x t f(t) dt}{(\int_0^x f(t) dt)^2}$$

$x=0$ 时

$$\lim_{x \rightarrow 0} \frac{g(x) - g(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{\int_0^x t f(t) dt}{x \int_0^x f(t) dt} \stackrel{\text{洛}}{=} \lim_{x \rightarrow 0} \frac{x f(x)}{\int_0^x f(t) dt + x f(x)}$$

$$\stackrel{\text{洛}}{=} \lim_{x \rightarrow 0} \frac{f(x) + x f'(x)}{2 f(x) + x f'(x)} \lim_{x \rightarrow 0} \frac{1}{1 + \frac{\int_0^x f(t) dt}{x f(x)}}$$

$$\lim_{x \rightarrow 0} \frac{\int_0^x f(t) dt}{x f(x)} = \lim_{x \rightarrow 0} \frac{\int_0^x f(t) dt}{x} \cdot \lim_{x \rightarrow 0} \frac{1}{f(x)} = 1$$

$$\therefore g'(x) = \begin{cases} \frac{f(x) \int_0^x f(t) dt - f(x) \int_0^x t f(t) dt}{(\int_0^x f(t) dt)^2}, & x \neq 0 \\ \frac{1}{2}, & x = 0 \end{cases}$$

例4.8

$$\text{设 } f(x) = \begin{cases} x^\alpha \sin x^\beta, & x > 0 \\ 0, & x \leq 0 \end{cases} \quad \beta < 0$$

(1) 在什么情况下, $f(x)$ 不是连续函数.

不连续 $\lim_{x \rightarrow 0^+} x^\alpha \sin x^\beta = 0$ 时连续
 $\alpha \leq 0$ 时 $f(x)$ 不连续

(2) $\alpha > 0$ 时 $f(x)$ 连续

$$\text{连续但不可导} \quad f(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{x^\alpha \sin x^\beta}{x} = \lim_{x \rightarrow 0} x^{\alpha-1} \sin x^\beta$$

$\alpha - 1 > 0$ 存在极限

$\therefore \alpha \in (0, 1]$ 连续但不可导

(3) $\alpha > 1$ 时可导

$$\begin{aligned} \text{可导但有界} \quad x > 0, f'(x) &= \alpha x^{\alpha-1} \sin x^\beta + x^\alpha \cos x^\beta \cdot \beta x^{\beta-1} \\ x < 0, f'(x) &= 0 \\ x = 0, f'(x) &= \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = 0 \quad (\alpha > 1) \end{aligned}$$

$\alpha + \beta - 1 < 0$ 时 $\rightarrow$ 无界振荡 $\rightarrow$ 无界

$$\begin{aligned} \text{可导有界但导函数不连续} \quad \lim_{x \rightarrow 0^+} f'(x) &= \lim_{x \rightarrow 0^+} (\alpha x^{\alpha-1} \sin x^\beta + \beta x^{\alpha+\beta-1} \cos x^\beta) \\ \alpha + \beta - 1 &\geq 0 \end{aligned}$$

(5) 导函数连续

$\Rightarrow \alpha + \beta - 1 > 0$

$\alpha + \beta - 1 = 0$ 时, $\exists$ 不连续

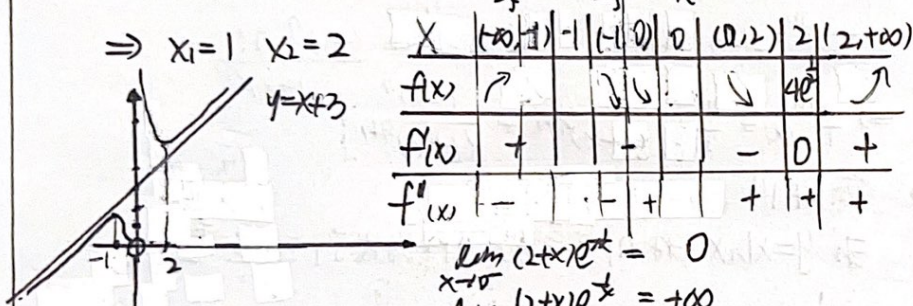
例 30. 1.5.11 ★ [大组长出的题]

$f(x) = (2+x)e^{\frac{1}{x}}$ 图像:

定义域: $(-\infty, 0) \cup (0, +\infty)$

$$f'(x) = e^{\frac{1}{x}} + (2+x)e^{\frac{1}{x}} \cdot \left(-\frac{1}{x^2}\right) = e^{\frac{1}{x}} \frac{(x-2)(x+1)}{x^2} \stackrel{?}{=} 0.$$

$$\Rightarrow x_1 = 1 \quad x_2 = 2$$



x	$(-\infty, -1)$	-1	$(-1, 0)$	0	$(0, 1)$	1	$(1, 2)$	2	$(2, +\infty)$
$f(x)$	$\nearrow$		$\searrow$		$\searrow$	$4e$	$\nearrow$		
$f'(x)$	$+$		$-$		$-$	0	$+$		
$f''(x)$	$-$		$+$		$+$	$+$	$+$		

$$\lim_{x \rightarrow 0^-} (2+x)e^{\frac{1}{x}} = 0$$

$$\lim_{x \rightarrow 0^+} (2+x)e^{\frac{1}{x}} = +\infty$$

$$f''(x) = \frac{(x+2)}{x^4} e^{\frac{1}{x}} \stackrel{?}{=} 0 \Rightarrow x = -\frac{2}{f} \quad f''(x) \text{ 不存在: } x=0.$$

$$\lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow +\infty} \frac{e^{\frac{1}{x}}}{\frac{1}{2+x}} = +\infty.$$

$$\lim_{x \rightarrow +\infty} \frac{f(x)}{x} = \lim_{x \rightarrow +\infty} \frac{f'(x)}{1} = 1$$

$$\lim_{x \rightarrow +\infty} [f(x) - x] = \lim_{x \rightarrow +\infty} (2+x)e^{\frac{1}{x}} - x = \lim_{x \rightarrow +\infty} \frac{(1+\frac{2}{x})e^{\frac{1}{x}} - 1}{\frac{1}{x}}$$

$$\stackrel{?}{=} \lim_{x \rightarrow +\infty} \frac{e^{\frac{1}{x}}}{\frac{1}{x^2}} \left(3+\frac{2}{x}\right) \stackrel{?}{=} \lim_{t \rightarrow 0} e^t (3+2t) = 3.$$

例 4.11.

设 $\begin{cases} x = 1+2t^2 \\ y = \int_1^{1+2\ln t} \frac{e^u}{u} du, (t>1), \end{cases}$ 求 $\frac{d^2y}{dx^2} \Big|_{x=9}$

[分析] $\frac{dx}{dt} = 4t, \frac{dy}{dt} = \frac{e^{1+2\ln t}}{1+2\ln t} \cdot 2 = \frac{2et}{(1+2\ln t)}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{e}{2(1+2\ln t)} = u(t)$$

$$\frac{d^2y}{dx^2} = \frac{u'(t)}{4t} = \frac{\frac{-2e \cdot \frac{2}{t}}{4(1+2\ln t)^2}}{4t} = \frac{-4e}{4t^2(1+2\ln t)^2}$$

$$x=9 \rightarrow t=2$$

$$\Rightarrow \frac{d^2y}{dx^2} \Big|_{x=9} = \frac{-e}{4 \times 4 (1+2\ln 2)^2} = \frac{-e}{16(1+2\ln 2)^2}$$

例 4.12 ★★

$$\begin{cases} x^x + tx - t^2 = 0 \\ \arctan(ty) = \ln(1+t^2y^2) \end{cases} \rightarrow \text{求 } \frac{dy}{dx}.$$

[分析]

$$x^x(\ln x + 1) \cdot x'_t + x + tx'_t - 2t = 0.$$

$$\Rightarrow x'_t = \frac{2t - x}{t + x^x(1+\ln t)}$$

$$\frac{1}{1+t^2y^2} \cdot (y + ty'_t) = \frac{2ty'_t + 2ty^2 \cdot y'_t}{1+t^2y^2}$$

$$\Rightarrow y'_t = \frac{y - 2ty^2}{2ty^2 - t} = \frac{y(1-2ty)}{-t(1-2ty)} = -\frac{y}{t}$$

$$\frac{dy}{dx} = \frac{(y - 2ty^2)(t + x^x(1+\ln t))}{(2ty^2 - t)(2t - x)} = \frac{y(t - x)(\ln x + 1) + 1}{x - 2t}$$

例 4.9

设 $y = \underbrace{\frac{x^3}{1-x}}_{y_1} \cdot \underbrace{\sqrt{\frac{2-x}{2+x}}}_{y_2} + e^{4x}$ 求 y'

解 $\ln|y_1| = 3\ln|x| - \ln|1-x| + \frac{1}{2}(\ln|2-x| - \ln|2+x|)$

$$\frac{y_1'}{y_1} = \frac{3}{x} + \frac{1}{1-x} + \frac{1}{2}\left(\frac{-1}{2-x} - \frac{1}{2+x}\right)$$

场合: 1) 大题一个步骤, 2) 最大似然估计.

例 4.10

设 $f(x) = \begin{cases} x^{3x}, & x > 0 \\ x+1, & x \leq 0 \end{cases}$ 求 $f''(x)$ $\lim_{x \rightarrow 0} \frac{\ln x}{\frac{1}{3x}} = \frac{\frac{1}{x}}{-\frac{1}{3x^2}} = -\frac{1}{3x}$

1) $f(0) = 1, \lim_{x \rightarrow 0^+} f(x) = 1 \Rightarrow f(x)$ 连续

2) $\lim_{x \rightarrow 0^+} \frac{f(x)-1}{x-0} = \lim_{x \rightarrow 0^+} \frac{x^{3x}-1}{x} = \lim_{x \rightarrow 0^+} \frac{e^{3x \ln x} - 1}{x} = \lim_{x \rightarrow 0^+} \frac{3x \ln x}{x} = \lim_{x \rightarrow 0^+} 3 \ln x = -\infty$

$\Rightarrow f'(0)$ 不存在 $\Rightarrow f''(0)$ 不存在

例 4.13

已知 $f(x) = \frac{x^2}{1-x^2}$, 求 $f^{(n)}(x)$

$f(x) = \frac{x^2-1+1}{1-x^2} = \frac{1}{1-x^2} - 1 = -1 + \frac{2}{(1-x)(1+x)}$
 $= -1 + \frac{1}{2}\left(\frac{1}{1-x} + \frac{1}{1+x}\right)$ 先化简

$\left(\frac{1}{1-x}\right)' = -\frac{-1}{(1-x)^2} = \frac{1}{(1-x)^2}$

$\left(\frac{1}{1-x}\right)'' = -2 \frac{-1}{(1-x)^3} = \frac{2}{(1-x)^3}$

$\left(\frac{1}{1-x}\right)^{(n)} = \frac{n!}{(1-x)^n}$

$\left(\frac{1}{1+x}\right)' = \frac{-1}{(1+x)^2} = -(1+x)^{-2}$

$\left(\frac{1}{1+x}\right)'' = -(-2)(1+x)^{-3} \dots \left(\frac{1}{1+x}\right)^{(n)} = \frac{(-1)^n n!}{(1+x)^n}$

$\Rightarrow f^{(n)}(x) = \frac{n!}{2} \left[\frac{1}{(1-x)^{n+1}} + \frac{(-1)^n}{(1+x)^{n+1}} \right]$

例 4.14

求 $y = x \ln x$ 的 n 阶导数, n 为大于 1 的正整数

$y' = \ln x + 1$

$y'' = \frac{1}{x} = x^{-1}$

$y^{(3)} = -1 \cdot x^{-2}$

$y^{(4)} = (-1)(-2)x^{-3} \Rightarrow y^{(n)} = (-1)^{n-2} x^{-(n-1)}, n \geq 2$

例 4.15

设 $f(x) = \arctan x$ 求 $f^{(n)}(0)$.

[分析] $f(x) = (\arctan x)' = \frac{1}{1+x^2}$

$\Rightarrow f'(x) \cdot (1+x^2) = 1$ 两边求 $n-1$ 阶导.

$C_{n-1}^{0} [f'(x)]^{(n-1)} \cdot (1+x^2) + C_{n-1}^1 [f'(x)]^{(n-2)} \cdot (2x) + C_{n-1}^2 [f'(x)]^{(n-3)} \cdot 2 = 0$

$[f^{(n)}(x)(1+x^2) + n f^{(n-1)}(x)x] = -(n-1)(n-2) f^{(n-2)}(x)$

$\Rightarrow f^{(n)}(0) = -(n-1)(n-2) f^{(n-2)}(0)$

$$f^{(0)}(0) = \arctan 0 = 0 \Rightarrow f^{(2k)}(0) = 0.$$

$$f^{(1)}(0) = 1, f^{(3)}(0) = -2 \cdot 1 \cdot f^{(1)}(0) = -2 \cdot 1$$

$$f^{(5)}(0) = -4 \cdot 3 \cdot f^{(3)}(0) = (-4) \cdot 3 \cdot (-2) \cdot 1$$

$$f^{(2k+1)}(0) = (-1)^k (2k)! \quad k=0, 1, 2, \dots$$

eg. $f(x) = \arctan x$, 则 $f^{(4)}(0) = ?$ $f^{(4)}(0) = 0.$

奇 $\rightarrow$ 偶 $\rightarrow$ 奇 $\rightarrow$ 偶 $\rightarrow$ 奇

例 4.16 $\frac{\sin x}{x}, x \neq 0$ 在点 $x=0$ 处的 18 阶导数 $f^{(18)}(0)$
 $f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

$$\sin x \quad f(x) = \sum_{n=0}^{\infty} \frac{f^{(2n)}(0)}{(2n)!} x^{2n} \quad (\text{抽象}).$$

$$f(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n+1)!} \quad (\text{具体}).$$

$$\Rightarrow f^{(2n)}(0) = \frac{(-1)^n (2n)!}{(2n+1)!} \Rightarrow f^{(18)}(0) = \frac{-18!}{18! \cdot 19!} = -\frac{1}{19}$$

例 4.17

$$f(x) = (x-1)^5 e^{-x} \text{ 求 } f^{(10)}(1).$$

$$f^{(10)}(1) = \sum_{m=0}^{\infty} \frac{f^{(m)}(1)}{m!} (x-1)^m \quad \text{抽象} \quad \boxed{\text{提出 } x-1}$$

具体: $f(x) = e(x-1)^5 e^{1-x}$
 $= \sum_{n=0}^{\infty} \frac{e^{-1} (-1)^n (x-1)^{n+5}}{n!} \rightarrow e^{-1} = \sum_{n=0}^{\infty} \frac{(-1)^n (x-1)^n}{n!}$

$$m=10, n=5:$$

$$\frac{f^{(10)}(1)}{10!} \cdot (x-1)^{10} = \frac{e^{-1} (-1)^5 (x-1)^{10}}{5!} \Rightarrow f^{(10)}(1) = -e^{-1} \frac{10!}{5!}$$

$\Rightarrow 4.1$

$$y = \arctan \frac{x-1}{x+1}$$

$$y' = \frac{1}{1 + \left(\frac{x-1}{x+1}\right)^2} \cdot \frac{(x+1) - (x-1)}{(x+1)^2} = \frac{1}{1+x^2} \quad \text{有.}$$

重点: 4.3 4.4 4.7 4.8 4.9

例 4.2

已知 $y = 1 + xe^{xy}$ 求 $y'(0)$, $y''(0)$.

[solution]

$$y' = e^{xy} + xe^{xy}(y + xy') = e^{xy}(xy + 1) + x^2 y' e^{xy}$$

$$y'(0) = 1 + 0 = 1. \quad e^{xy}(y + xy') +$$

$$y'' = e^{xy}(xy + 1)(y + xy') + (2xy' + x^2 y'') e^{xy} + (x^2 y' e^{xy})(y + xy')$$

$$y''(0) = 1 + 1 + 0 + 0 = 2$$

例 4.3

$$f(x) = \int_{-1}^x \sqrt{1-e^t} dt. \quad x = f^{-1}(y) \quad \frac{dx}{dy} \Big|_{y=0} = ?$$

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{\sqrt{1-e^x}} \Big|_{y=0, x=-1} = \frac{1}{\sqrt{1-e^{-1}}}$$

例 4.4

$$f(x) = \begin{cases} x^2 e^{-x^2}, & |x| \leq 1 \\ \frac{1}{e}, & |x| > 1 \end{cases} \quad f'(x) = ?$$

$$f'(x) = 0, \quad x < -1 \text{ 或 } (x > 1)$$

$$f'(x) = 2xe^{-x^2} + x^2 e^{-x^2}(-2x) = 2xe^{-x^2}(1-x^2)$$

$$\lim_{x \rightarrow 0} f'(x) = 0 \quad (x=1)$$

例 4.5

$$y = [(1+x)(3+x)^9]^{\frac{1}{2}} (2+x)^4. \quad y'(0) = ?$$

$$y' = [(1+x)(3+x)^9]^{\frac{1}{2}} 4(2+x)^3 + (2+x)^4 \frac{1}{2} [(1+x)(3+x)^9]^{-\frac{1}{2}} \cdot$$

$$\cdot [(3+x)^9 + (1+x)^9 (3+x)^8]$$

$$y'(0) = [3^9]^{\frac{1}{2}} \cdot 4 \cdot 2^3 + 2^4 \cdot \frac{1}{2} [3^9]^{-\frac{1}{2}} \cdot [3^9 + 9 \cdot 3^8]$$

$$= [3^9]^{\frac{1}{2}} \cdot 2^5 + 2^3 \cdot [3^9]^{-\frac{1}{2}} \cdot 3^9$$

$$= 2^6 [3^9]^{\frac{1}{2}}$$

例 4.6

$$\begin{cases} x = t + e^t \\ y = \sin t \end{cases} \quad \frac{d^2 y}{dx^2} \Big|_{t=0} = ?$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\cos t}{1+e^t} = u.$$

$$\frac{d^2 y}{dx^2} = \frac{du}{dx} = \frac{du/dt}{dx/dt} = \frac{\frac{-\sin t(1+e^t) - \cos t(e^t)}{(1+e^t)^2}}{1+e^t}$$

$$= \frac{-\sin t(1+e^t) - e^t \cos t}{(1+e^t)^3} \Big|_{t=0} = \frac{-1}{2^3}$$

例 4.7

$$y = \frac{5}{2+3x-2x^2} \quad \text{求 } n \text{ 阶导数.}$$

$$y = \frac{5}{(2x+1)(-x+2)} = \frac{A}{2x+1} - \frac{B}{x-2} = \frac{-2}{2x+1} + \frac{1}{x-2}$$

$$\begin{aligned} A - 2B &= 0, & -2A - B &= 5 \\ A &= 2B, & B &= -1, & A &= -2 \end{aligned}$$

$$y^{(n)} = (-1)^n n! [(x-2)^{-n-1} - 2(2x+1)^{-n-1}]$$

例 4.8

$y = (\arcsin x)^2$, 证明:

$$(1-x^2)y^{(n+2)} - (2n+1)xy^{(n+1)} - n^2y^{(n)} = 0.$$

求 $y'(0), y''(0), \dots, y^{(n)}(0)$.

$$y = (\arcsin x)^2$$

$$y' = 2(\arcsin x) \cdot \frac{1}{\sqrt{1-x^2}} \quad (1-x^2)^{-\frac{1}{2}}$$

$$y'' = 2 \frac{1}{1-x^2} + 2 \arcsin x \left(-\frac{1}{2}\right) (1-x^2)^{-\frac{3}{2}} \cdot 2x$$

$$(1-x^2)y'' = 2 - 2x \arcsin x (1-x^2)^{-\frac{1}{2}} \Rightarrow 2 - xy' = (1-x^2)y''$$

两边求 n 阶导数.

$$C_n^0 (1-x^2)y^{(n+2)} + C_n^1 (-2x)y^{(n+1)} + C_n^2 (2x^2)y^{(n)} = xy^{(n+1)} + C_n^1 y^{(n)}$$

$$\Rightarrow (1-x^2)y^{(n+2)} - (2n+1)xy^{(n+1)} - n^2y^{(n)} = 0.$$

$$y'(0) = 0$$

$$y''(0) = 2$$

$$y^{(2k+1)}(0) = 0$$

$$y^{(n+2)}(0) = n^2 y^{(n)}(0)$$

$$y^{(2k)}(0) = 2^{k-1} [(k-1)!]^2$$

例 4.9

函数 $y = \ln(1-2x)$ 在 $x=0$ 处 n 阶导数 $y^{(n)}(0) = ?$

$$y' = (1-2x)^{-1} (-2).$$

$$y'' = (-1)(1-2x)^{-2} (-2)^2$$

$$y^{(n)} = \frac{(-1)^{n-1} (n-1)!}{(1-2x)^{-n}} (-2)^n = (-1)^{n-1} (n-1)! (1-2x)^{-n} 2^n$$

$$y^{(n)}(0) = - (n-1)! (1-2x)^{-n} \cdot 2^n$$

$$y^{(n)}(0) = - (n-1)! 2^n.$$

1000.4.2

设 $f(x) = x^2 e^x$ 则 $n > 1$, $f(x)$ 在 $x=0$ 处的 n 阶导数 $f^{(n)}(0)$ [★ 泰勒展开 + 泰勒展开 $\Rightarrow$ 系数对比].

$$1). x^2 = e^{x \ln 2} = 1 + x \ln 2 + \dots + \frac{(x \ln 2)^{n-2}}{(n-2)!} + o(x^{n-2})$$

$$x^2 e^x = x^2 + x^3 \ln 2 + \dots + \frac{(\ln 2)^{n-2}}{(n-2)!} \cdot x^n + o(x^n)$$

$$2). f(x) = \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} (x-0)^k + o(x^n)$$

$$3). f(x) = \text{系数比较} \quad \frac{f^{(n)}(0)}{n!} = \frac{(\ln 2)^{n-2}}{(n-2)!}$$

$$\Rightarrow f^{(n)}(0) = n(n-1)(\ln 2)^{n-2}$$

1000. 4.10

$$\text{设 } f(x) = \frac{x}{1-2x^4} \quad f^{(101)}(0) = ?$$

[solution]

$$\begin{aligned} f(x) &= x \sum_{n=0}^{\infty} (2x^4)^n = \sum_{n=0}^{\infty} 2x^{4n+1} \\ &= \sum_{n=0}^{\infty} 2^n x^{4n+1} \Rightarrow f^{(101)}(x) = 2^{25} x^{101} \end{aligned}$$

$$f(x) = x \sum_{n=0}^{\infty} (2x^4)^n = \sum_{n=0}^{\infty} 2^n x^{4n+1} \quad 4n+1=101 \Rightarrow n=25.$$

$$f^{(101)}(x) = 2^{25} \cdot (101)!.$$

1000. 4.13

$$y = f\left(\frac{3x-2}{3x+2}\right) \text{ 且 } f'(x) = \arctan x^2 \quad \frac{dy}{dx}\bigg|_{y=0}.$$

$$y = f(u) \quad u = g(x)$$

$$\begin{aligned} \frac{dy}{dx} &= \arctan(u^2) \cdot u' \\ &= \left[\arctan\left(\frac{3x-2}{3x+2}\right)^2 \right] \cdot \frac{3(3x+2) - (3x-2)3}{(3x+2)^2} \end{aligned}$$

$$x=0 \Rightarrow \frac{0}{4} \cdot 3 = \frac{3}{4} \pi.$$

1000. 4.15.

$$\begin{cases} x = 1+t^2 \\ y = \cos t \end{cases} \rightarrow \frac{dy}{dx} \text{ 和 } \frac{d^2y}{dx^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-\sin t}{2t}, \quad \frac{d^2y}{dx^2} = \frac{\sin t - t \cos t}{4t^3} \quad \text{求 } \lim_{x \rightarrow 1^+} \frac{d^2y}{dx^2}$$

$$\lim_{x \rightarrow 1^+} \frac{d^2y}{dx^2} = \lim_{t \rightarrow 0^+} \frac{\sin t - t \cos t}{4t^3} = \lim_{t \rightarrow 0^+} \frac{\cos t - \cos t + t \sin t}{12t^2} = \frac{1}{12}$$

1000. 4.16.

$$f(x) = \arctan x, \quad f'(0)=1, \quad f''(0)=2$$

$$\begin{cases} x = f(t) - \pi \\ y = f(e^{3t} - 1) \end{cases} \quad \text{求 } \frac{dy}{dx}\bigg|_{t=0}, \quad \frac{d^2y}{dx^2}\bigg|_{t=0}.$$

[solution].

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{f'(e^{3t}-1) \cdot e^{3t} \cdot 3}{f'(t)} \bigg|_{t=0} = \frac{1 \cdot 1 \cdot 3}{1} = 3.$$

$$\frac{d^2y}{dx^2} = \frac{[f''(e^{3t}-1) \cdot (3e^{3t})^2 + f'(e^{3t}-1) \cdot 9e^{3t}] f'(t) - [f'(e^{3t}-1) \cdot e^{3t} \cdot 3] f''(t)}{[f'(t)]^3}$$

$$\begin{aligned} t=0 &= \frac{[f''(0) \cdot 9 + f'(0) \cdot 9] f'(0) - [f'(0) \cdot 3] \cdot f''(0)}{[f'(0)]^3} \\ &= \frac{27 - 6}{1} = 21 \end{aligned}$$

1000. 4.21.

$$\ln(x^2+4) = x^3 y + \sin y x \quad \text{求 } dy|_{x=0}.$$

[注意: 问的是 dy 不是 $\frac{dy}{dx}$]

$$x=0 \Rightarrow y=1$$

$$\text{求 } dy = dx$$

1000. 4t. 4

设 $f(x)$ 可微, 证明 若 $x=1$ 时 $\frac{df(x)}{dx} = \frac{df^2(x)}{dx}$

则 $f'(1)=0$. 或 $f(1)=1$

[solution]. $\frac{df(x)}{dx} = \frac{1}{2x} \frac{df(x)}{du} \Rightarrow \frac{df(x)}{dx} \Big|_{x=1} = 0$
 $\frac{df^2(x)}{dx} = 2f(x) \frac{df(x)}{dx}$ 或 $\frac{1}{2x} = 2f(x) \Rightarrow f(x) = \frac{1}{4x}$

1000. 4t. 5

设 $f(x) = \lim_{n \rightarrow \infty} x \cos 2x \cos \frac{x}{2} \cos \frac{x}{4} \cdots \cos \frac{x}{2^n} \quad (x > 0)$

(1). 证明 $f(x) = \cos 2x \sin x$

或 求 $f^{(20)}(x)$

[solution].

$$\begin{aligned} & \lim_{n \rightarrow \infty} \frac{x}{\sin \frac{x}{2^n}} \cos 2x \cos \frac{x}{2} \cos \frac{x}{4} \cdots \left(\cos \frac{x}{2^n} \cdot \sin \frac{x}{2^n} \right) \\ &= \lim_{n \rightarrow \infty} \frac{x}{\sin \frac{x}{2^n}} \cos 2x \cos \frac{x}{2} \cos \frac{x}{4} \cdots \cos \frac{x}{2^{n-1}} \sin \frac{x}{2^{n-1}} \cdot \frac{1}{2} \\ &= \lim_{n \rightarrow \infty} \frac{(x)}{\sin \frac{(x)}{2^n}} \cos 2x \cos \frac{x}{2} \cdots \sin x \left(\frac{1}{2} \right)^n = \cos 2x \sin x. \end{aligned}$$

(2). $f^{(20)}(x) = [\sin x - 2 \sin^3 x]^{(20)} = \frac{1}{2} [\sin 3x - \sin x]^{(20)}$

$$\begin{aligned} &= \sin \left(x + \frac{20}{2} \pi \right) - \sin \left(x + \frac{20}{2} \pi \right) = \frac{1}{2} [3^{20} \sin(2x + \frac{20}{2} \pi) - \sin(x + \frac{20}{2} \pi)] \\ &= \sin x = \frac{1}{2} [3^{20} \sin 3x - \sin x] \end{aligned}$$

1000. 4t. 8.

设 n 为正整数, $f(x) = \ln(\sqrt{1+x^2} - x)$ 求 $f^{(2n+1)}(0)$

1). $f(x) = \frac{-1}{\sqrt{1+x^2}} = -(1+x^2)^{-\frac{1}{2}}$

$f'(x) = (-1)(-\frac{1}{2})(1+x^2)^{-\frac{3}{2}}(2x) = \frac{x}{(1+x^2)^{\frac{3}{2}}}$

$\Rightarrow (1+x^2)f'(x) + x f'(x) = 0$

用莱布尼兹法则: 求 $n-1$ 阶导.

$$\begin{aligned} & C_{n-1}^0 (1+x^2) f^{(n+1)}(x) + C_{n-1}^1 (2x) f^{(n)}(x) + C_{n-1}^2 2 f^{(n-1)}(x) \\ &+ C_{n-1}^3 x^2 f^{(n-2)}(x) + \cdots + C_{n-1}^{n-1} f^{(1)}(x) = 0. \end{aligned}$$

$x=0 \Rightarrow f^{(n+1)}(0) + [2C_{n-1}^1 + (n-1)] f^{(n-1)}(0) = 0$

$f^{(n+1)}(x) + (n-1)^2 f^{(n-1)}(x) = 0$

$n=0 \quad f'(x)|_{x=0} = -1$

$n=1 \quad f^{(3)}(x)|_{x=0} = -2^2 \cdot (-1)$

$n=2 \quad f^{(5)}(x)|_{x=0} = (-4^2)(-2^2)(-1^2)$

$n=n \quad f^{(2n+1)}(x)|_{x=0} = (-1)^{n+1} [(2n-1)!!]^2$

$f^{(n+1)}(x) = -(n-1)^2 f^{(n-1)}(x)$
 $n=1, 3, 5, 7, \dots$
 $0 \quad 1 \quad 2 \quad 3 \dots$